Sommersemester 2026

1. Exercise Sheet Solutions

TUTORIUM: Friday, 20.03.2026.

SECTION

01

Getting familiar with the Density of States

The calculation of thermodynamic quantities, response functions and Feynman diagrams in QFT for condensed matter systems often requires the evaluation of integrals or sums over all momenta $\boldsymbol{k}$ (typically over the first Brillouin Zone). An important simplification of these $\boldsymbol{k}$-summations is possible, however, when the integrand $\mathcal{F}$ depends on the **energy only**. In this case the integration/sum is best performed by using the energy ε as a variable. In the case of a cubic lattice of volume L^d in d dimensions, for a given observable F, we have:

$$F = \frac{1}{L^d}\sum_{\boldsymbol{k}} \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \frac{1}{(2\pi)^d}\frac{(2\pi)^d}{L^d}\sum_{\boldsymbol{k}} \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \int \mathrm{d}\varepsilon\, \mathcal{N}(\varepsilon)\mathcal{F}(\varepsilon)$$

or, for the continuous case,

$$F = \frac{1}{(2\pi)^d}\int \mathrm{d}^d k\, \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \int \mathrm{d}\varepsilon\, \mathcal{N}(\varepsilon)\mathcal{F}(\varepsilon),$$

where $\mathcal{N}(\varepsilon)$, i.e. the so-called Density of States (DOS), can be defined by comparison of the different expressions as

$$\mathcal{N}(\varepsilon) = \frac{1}{L^d}\sum_{\boldsymbol{k}} \delta(\varepsilon - \varepsilon_{\boldsymbol{k}})$$

or, for the continuous case,

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d}\int \mathrm{d}^d k\, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

a)

PROBLEM

Consider the two cases of particles which can move freely and particles whose motion is bound to an infinite lattice with lattice spacing a. Which of the above expressions do you have to use in the first and second case, respectively? Is the integral/summation restricted to certain $\boldsymbol{k}$-vectors? How does the result change, if one considers a one-dimensional, finite lattice (N lattice points, lattice spacing a) with periodic boundary conditions?

SOLUTION

Freely moving particles can have arbitrary momenta, requiring integration over all of space

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Particles bound to an infinite lattice with lattice spacing a can only have momenta within the first Brillouin zone, i.e.

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{[0, \frac{2\pi}{a}]^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Finally, for a one-dimensional, finite (N lattice points, spacing a) lattice with periodic boundary conditions, the summation is restricted to a discrete set of allowed momenta

$$\mathcal{N}(\varepsilon) = \frac{1}{N} \sum_{k \in K} \delta(\varepsilon - \varepsilon_k)$$

with $K = \left\{ \frac{2\pi}{a} \frac{j}{N} \mid j = 0, \ldots, N-1 \right\}$.

b)

PROBLEM

Calculate and plot the explicit expression for $\mathcal{N}(\varepsilon)$ for free, non-interacting particles of mass m (so that $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}$) in one, two and three dimensions. How do the corresponding Fermi surfaces look like in these cases?

SOLUTION

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}})$$

with $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}$. We have

$$\int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(g(\boldsymbol{k})) = \int_{g^{-1}(0)} \frac{1}{|\boldsymbol{\nabla} g|} \, \mathrm{d}\sigma(k)),$$

which for $g(\boldsymbol{k}) = \varepsilon - \varepsilon_{\boldsymbol{k}}$, $g^{-1}(0) = S^{d-1}(k_\mathrm{F})$ and $|\boldsymbol{\nabla} g| = \frac{\hbar^2 k}{m}$ yields

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \left| S^{d-1}(k_\mathrm{F}) \right| \frac{1}{\frac{\hbar^2 k_\mathrm{F}}{m}}$$

where $k_\mathrm{F} = \frac{\sqrt{2m\varepsilon}}{\hbar}$ is the Fermi momentum and $S^{d-1}(k)$ is the radius-k sphere in d dimensions.

- $d = 1$: $|S^0(k)| = 2$, hence

$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi}\frac{\sqrt{2m}}{\hbar\sqrt{\varepsilon}}$$

Fermi surface: two points at $k = \pm k_{\mathrm{F}}$.

- $d = 2$: $|S^1(k)| = 2\pi k$, hence

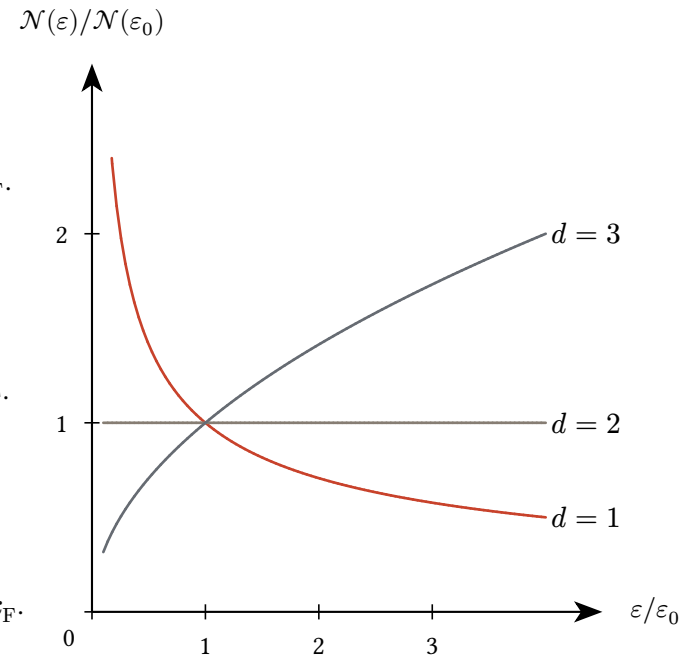
$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi}\frac{m}{\hbar^2}$$

Fermi surface: circle with radius k_{F}.

- $d = 3$: $|S^2(k)| = 4\pi k^2$, hence

$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi^2}\frac{m}{\hbar^2}\frac{\sqrt{2m\varepsilon}}{\hbar}$$

Fermi surface: sphere with radius k_{F}.



c)

PROBLEM

Consider the following one-dimensional tight-binding Hamiltonian

$$H = -t \sum_{\langle i,j\rangle,\sigma} \left(c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}\right).$$

Here the hopping ($t > 0$) is restricted to neighboring sites, where $c^\dagger_{i,\sigma}$ and $c_{i,\sigma}$ are the creation/annihilation operators for one electron with spin $\sigma = \uparrow, \downarrow$ at the position $x_i = ia$ with $i = 0, 1, ..., N-1$ and a being the lattice spacing. Assuming periodic boundary conditions ($x_0 = x_N$), compute the corresponding eigenenergies, e.g. using the following basis transformation (from real to momentum space)

$$c^\dagger_{k,\sigma} = \frac{1}{\sqrt{N}} \sum_{x_i} e^{-ikx_i} c^\dagger_{i,\sigma}$$

for the fermionic operators.

SOLUTION

We use the given basis transformation

$$c^\dagger_{k,\sigma} = \frac{1}{\sqrt{N}} \sum_{x_i} e^{-ikx_i} c^\dagger_{i,\sigma}$$

and invert it to express the real-space operators in terms of the momentum-space operators

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}} \sum_{k} e^{ikx_i} c^\dagger_{k,\sigma}$$

$$c_{i,\sigma} = \frac{1}{\sqrt{N}} \sum_{k} e^{-ikx_i} c_{k,\sigma}.$$

Plugging into the Hammiltonian, we arrive at (here, $j+1$ is to be understood modulo N due to the periodic boundary conditions)

$$
\begin{aligned}
H &= -t \sum_{j,\sigma} \left(c^\dagger_{j,\sigma} c_{j+1,\sigma} + \text{h.c.} \right) \\
&= -t \sum_{j,\sigma} \left(\frac{1}{\sqrt{N}} \sum_k e^{ikx_j} c^\dagger_{k,\sigma} \frac{1}{\sqrt{N}} \sum_{k'} e^{-ik'x_{j+1}} c_{k',\sigma} + \text{h.c.} \right) \\
&= -t \frac{1}{N} \sum_{k,k'} \sum_{j,\sigma} \left(e^{ikja - ik'(j+1)a} c^\dagger_{k,\sigma} c_{k',\sigma} + \text{h.c.} \right)
\end{aligned}
$$

Now we use the identity

$$
\frac{1}{N} \sum_j e^{ikja - ik'(j+1)a} = e^{-ik'a} \frac{1}{N} \sum_j e^{ija(k-k')} = e^{-ik'a} \delta_{k,k'}
$$

to arrive at

$$
\begin{aligned}
H &= -t \sum_{k,k'} \sum_\sigma \left(e^{-ik'a} \delta_{k,k'} c^\dagger_{k,\sigma} c_{k',\sigma} + \text{h.c.} \right) \\
&= -t \sum_{k,\sigma} \left(e^{-ika} c^\dagger_{k,\sigma} c_{k,\sigma} + \text{h.c.} \right) \\
&= -t \sum_{k,\sigma} \left(e^{-ika} c^\dagger_{k,\sigma} c_{k,\sigma} + e^{ika} c^\dagger_{k,\sigma} c_{k,\sigma} \right) \\
&= \sum_{k,\sigma} \varepsilon_k c^\dagger_{k,\sigma} c_{k,\sigma}
\end{aligned}
$$

with eigenenergies

$$
\varepsilon_k = -2t \cos(ka)
$$

and eigenstates

$$
|(k,\sigma)\rangle = c^\dagger_{k,\sigma} |0\rangle.
$$

d)

PROBLEM

How can one extend the results of 1c) to arbitrary dimensions $d > 1$? Plot numerically the DOS $\mathcal{N}(\varepsilon)$ for the cases $d = 1, 2, 3$ with $\hbar = m = t = a = 1$. Which are the most prominent features of these DOS functions and at which energies ε do they occur? How would the corresponding Fermi surfaces look like for the cases $d = 1, 2$, e.g. if one has an average density of one electron per site (half-filled system)?

SOLUTION

In arbitrary dimension $d > 1$, the tight-binding Hamiltonian is

$$
H = \sum_{\boldsymbol{k},\sigma} \varepsilon_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k},\sigma} c_{\boldsymbol{k},\sigma},
$$

with dispersion

$$
\varepsilon_{\boldsymbol{k}} = -2t \sum_{i=1}^{d} \cos(k_i a).
$$

The density of states is therefore

$$\mathcal{N}_d(\varepsilon) = \frac{1}{(2\pi)^d} \int_{[0,2\frac{\pi}{a}]^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Using

$$\int_{[0,2\frac{\pi}{a}]^d} \mathrm{d}^d k \, \delta(g(\boldsymbol{k})) = \int_{g^{-1}(0)\cap[0,2\frac{\pi}{a}]^d} \frac{\mathrm{d}\sigma(\boldsymbol{k})}{|\boldsymbol{\nabla} g|},$$

with $g(\boldsymbol{k}) = \varepsilon - \varepsilon_{\boldsymbol{k}}$, and setting $\hbar = m = t = a = 1$, we obtain

$$\boldsymbol{\nabla} g = -2 \begin{pmatrix} \sin(k_1) \\ \vdots \\ \sin(k_d) \end{pmatrix}, \quad |\boldsymbol{\nabla} g| = 2\sqrt{\sum_{i=1}^{d} \sin^2(k_i)}.$$

- $d = 1$: For $|\varepsilon| < 2$, there are two Fermi points $\pm k_{\mathrm{F}}$ with

  $$\cos(k_{\mathrm{F}}) = -\frac{\varepsilon}{2}.$$

  Hence

  $$|\boldsymbol{\nabla} g| = 2|\sin(k_{\mathrm{F}})| = 2\sqrt{1 - \cos^2(k_{\mathrm{F}})} = \sqrt{4 - \varepsilon^2},$$

  and therefore

  $$\mathcal{N}_1(\varepsilon) = \begin{cases} \dfrac{1}{\pi\sqrt{4 - \varepsilon^2}} & \text{if } |\varepsilon| < 2 \\ 0 & \text{otherwise} \end{cases}.$$

- $d = 2$: We write

  $$\mathcal{N}_2(\varepsilon) = \frac{1}{(2\pi)^2} \int_0^{2\pi} \mathrm{d}k_x \int_0^{2\pi} \mathrm{d}k_y \, \delta\big(\varepsilon + 2\cos(k_x) + 2\cos(k_y)\big).$$

  For fixed k_x, define

  $$A = \varepsilon + 2\cos(k_x).$$

  Then the inner integral is exactly the $d = 1$ result:

  $$\int_0^{2\pi} \mathrm{d}k_y \, \delta\big(A + 2\cos(k_y)\big) = \frac{2}{\sqrt{4 - A^2}}.$$

  Thus

  $$\mathcal{N}_2(\varepsilon) = \frac{1}{(2\pi)^2} \int_0^{2\pi} \mathrm{d}k_x \, \frac{2}{\sqrt{4 - (\varepsilon + 2\cos(k_x))^2}}.$$

  With $u = \cos(k_x)$ and symmetry, this becomes

  $$\mathcal{N}_2(\varepsilon) = \frac{1}{\pi^2} \int_{-1}^{1} \frac{\mathrm{d}u}{\sqrt{1 - u^2}\sqrt{4 - (\varepsilon + 2u)^2}}.$$

  This is the complete elliptic integral of the first kind, so

$$\mathcal{N}_2(\varepsilon) = \begin{cases} \frac{1}{2\pi^2} K\left(\sqrt{1-\frac{\varepsilon^2}{16}}\right) & \text{if } |\varepsilon| < 4 \\ 0 & \text{otherwise} \end{cases}.$$

- $d = 3$: Starting from

$$\mathcal{N}_3(\varepsilon) = \frac{1}{(2\pi)^3} \int_0^{2\pi} \mathrm{d}k_x \int_0^{2\pi} \mathrm{d}k_y \int_0^{2\pi} \mathrm{d}k_z \, \delta\big(\varepsilon + 2\cos(k_x) + 2\cos(k_y) + 2\cos(k_z)\big),$$

we fix k_z and define

$$\varepsilon' = \varepsilon + 2\cos(k_z).$$

The remaining k_x, k_y integral is then exactly the $d = 2$ DOS at energy ε':

$$\mathcal{N}_3(\varepsilon) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}k_z \, \mathcal{N}_2(\varepsilon + 2\cos(k_z)).$$

Inserting the $d = 2$ result gives

$$\mathcal{N}_3(\varepsilon) = \frac{1}{4\pi^3} \int_0^{2\pi} \mathrm{d}k_z \, K\left(\sqrt{1 - \frac{(\varepsilon + 2\cos(k_z))^2}{16}}\right),$$

where the integrand contributes only if

$$|\varepsilon + 2\cos(k_z)| < 4.$$

Equivalently, with $u = \cos(k_z)$,

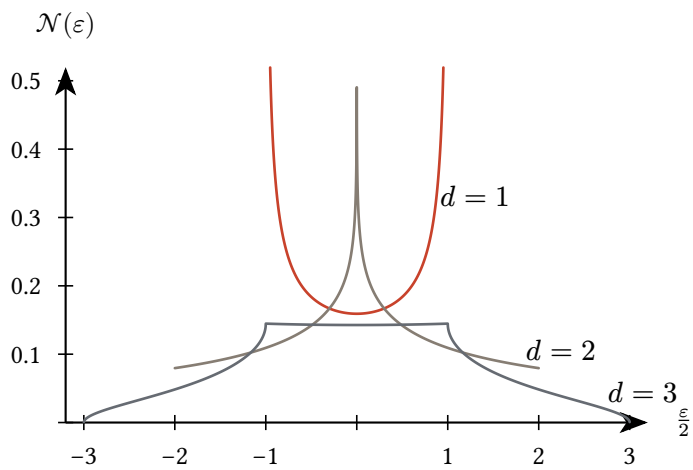
$$\mathcal{N}_3(\varepsilon) = \frac{1}{2\pi^3} \int_{-1}^{1} \frac{K\left(\sqrt{1 - \frac{(\varepsilon+2u)^2}{16}}\right)}{\sqrt{1-u^2}} \, \mathrm{d}u,$$

again with the restriction

$$|\varepsilon + 2u| < 4.$$

In particular,

$$\mathcal{N}_3(\varepsilon) = 0 \quad \text{for} \quad |\varepsilon| > 6.$$



SECTION

02

Screened and Unscreened Coulomb Potentials

a)

PROBLEM

From the integral representation of the delta function,

$$\delta(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$$

and the fact that the Coulomb potential $\phi(\boldsymbol{r}) = -e/r$ satisfies Poisson's equation,

$$-\nabla^2 \phi(\boldsymbol{r}) = -4\pi e \delta(\boldsymbol{r}),$$

show that the electronic pair potential, $V(\boldsymbol{r}) = -e\phi(\boldsymbol{r}) = e^2/r$, can be written in the form

$$V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}),$$

where the Fourier transform $V(\boldsymbol{k})$ is given by

$$V(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

SOLUTION

Assume the Coulomb potential admits the Fourier representation

$$\phi(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \phi(\boldsymbol{k}).$$

Applying $-\nabla^2$ gives

$$-\nabla^2 \phi(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} (-\nabla^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}}) \phi(\boldsymbol{k}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} k^2 \phi(\boldsymbol{k}),$$

since

$$-\nabla^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}} = k^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}}.$$

On the other hand, Poisson's equation gives

$$-\nabla^2 \phi(\boldsymbol{r}) = -4\pi e \delta(\boldsymbol{r}) = -4\pi e \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}.$$

Comparing the Fourier coefficients of $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, we obtain

$$k^2 \phi(\boldsymbol{k}) = -4\pi e,$$

hence

$$\phi(\boldsymbol{k}) = -\frac{4\pi e}{k^2}.$$

Therefore

$$V(\boldsymbol{k}) = -e\phi(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

Thus

$$V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}),$$

with

$$V(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

b)

PROBLEM

Show that the Fourier transform of the screened Coulomb interaction $V_s(\boldsymbol{r}) = (e^2/r)e^{-k_{\mathrm{TF}}r}$ is

$$V_s(\boldsymbol{k}) = 4\pi \frac{e^2}{k^2 + k_{\mathrm{TF}}^2}$$

by substituting this expression into the Fourier integral

$$V_s(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V_s(\boldsymbol{k}),$$

and evaluating that integral in spherical coordinates (Hint: The radial integral is best done as a contour integral.). Finally, deduce that $V_s(\boldsymbol{r})$ satisfies

$$(-\nabla^2 + k_{\mathrm{TF}}^2) V_s(\boldsymbol{r}) = 4\pi e^2 \delta(\boldsymbol{r}).$$

SOLUTION

Write your solun hee.